

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (Electrical Engineering)

December 2011

EE 205 Electrical Machines – I

Date: 06.12.2011, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Explain the working of transformer at load and no load conditions. [07]

Q - 2 (a) What are the different losses occur in a transformer? Derive the condition for maximum efficiency of the transformer. [06]

(b) Explain with reasons what happens when [03]

[1] The primary of a power transformer is connected to dc supply of the same voltage rating.

[2] The primary terminals of a single phase transformer are connected to rated supply while secondary terminals are short circuited.

(c) A 230/230 V, 3 KVA transformer gave the following results: [05]

O.C. Test: 230 V, 2 amp, 100W

S.C. Test: 15 V, 13 amp, 120W

Determine the regulation and efficiency at full load 0.80 p.f. lagging.

OR

Q - 2 (a) Explain scott connection of two transformers in brief. [07]

(b) A 30 kVA, 2400/120 V, 50 Hz transformer has a high voltage winding resistance of 0.1Ω and a leakage reactance of 0.22Ω . The low voltage winding resistance is 0.035Ω and the leakage reactance is 0.012Ω . Find the equivalent winding resistance, reactance and impedance referred to the (i) High voltage side and (ii) the low voltage side. [07]

Q - 3 Attempt any two: [14]

[1] Auto-Transformer

[2] Cooling of Transformer

[3] Three phase to two phase conversion of Transformers.

[4] Back to Back test of Transformer

SECTION – II

Q - 4 (a) What is electromechanical conversion device? Explain the working of a generator with the help of power flow diagram. [05]

- (b) Define the following terms: [02]

[1] Pole pitch [2] Coil span [3] Resultant pitch [4] Front pitch

- Q - 5 (a) Explain the various factors affecting the speed of dc motor and explain Ward – Leonard system used for the controlling the speed of DC shunt motor. [07]

- (b) Armature of a 6 – pole, 6 – circuit dc shunt motor takes 400 A at a speed of 350 rpm. The flux per pole is 80 mWb, the number of armature turns is 600 and 3% of the torque is lost in windage, friction and iron-loss. Calculate the brake horse power. [07]

OR

- Q - 5 (a) Derive an equation of generated emf of a generator. [05]

- (b) The following figures give the O.C.C. of a dc shunt generator at 300 rpm. [06]

Field Amperes	0	2	3	4	5	6	7
Armature Voltage	7.5	92	132	162	183	190	212

Plot the O.C.C for 375 rpm and determine the voltage to which the machine will excite if field circuit resistance is 40 Ω .

- [a] What additional resistance would have to be inserted in the field circuit to reduce the voltage to 200 V at 375 rpm?
- [b] Without this additional resistance, determine the load current supplied by the generator, when its terminal voltage is 200 V. Ignore armature reaction and assume speed to be constant. Armature resistance is 0.4 Ω .
- (c) A long-shunt generator delivers a load current of 50 A at 500 V and has armature, series field and shunt field resistances of 0.05 Ω , 0.03 Ω and 250 Ω respectively. Calculate the generated voltage and the armature current. Allow 1 V per brush for contact drop. [03]

- Q - 6 Attempt any two: [14]

- [1] Parallel operation of DC generators
[2] Characteristics of DC shunt motor
[3] Load sharing
[4] Electric braking of dc series motor.
